

PROJECT EXPO-2026

SMART WILD PIG DETECTION AND CROP PROTECTION USING IoT AND AI

Phase 3 - System Design Review, Deliverables

TEAM MEMBERS:

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❖ System Architecture:

- Overall Solution Architecture
- Data Flow Diagram
- Functional Block Diagram

- Hardware-Software Partitioning

❖ **Hardware Design:**

- Final Block Diagram
- Circuit Schematics
- Component Selection
- Bill of Materials (BOM)
- PCB Layout (if applicable)

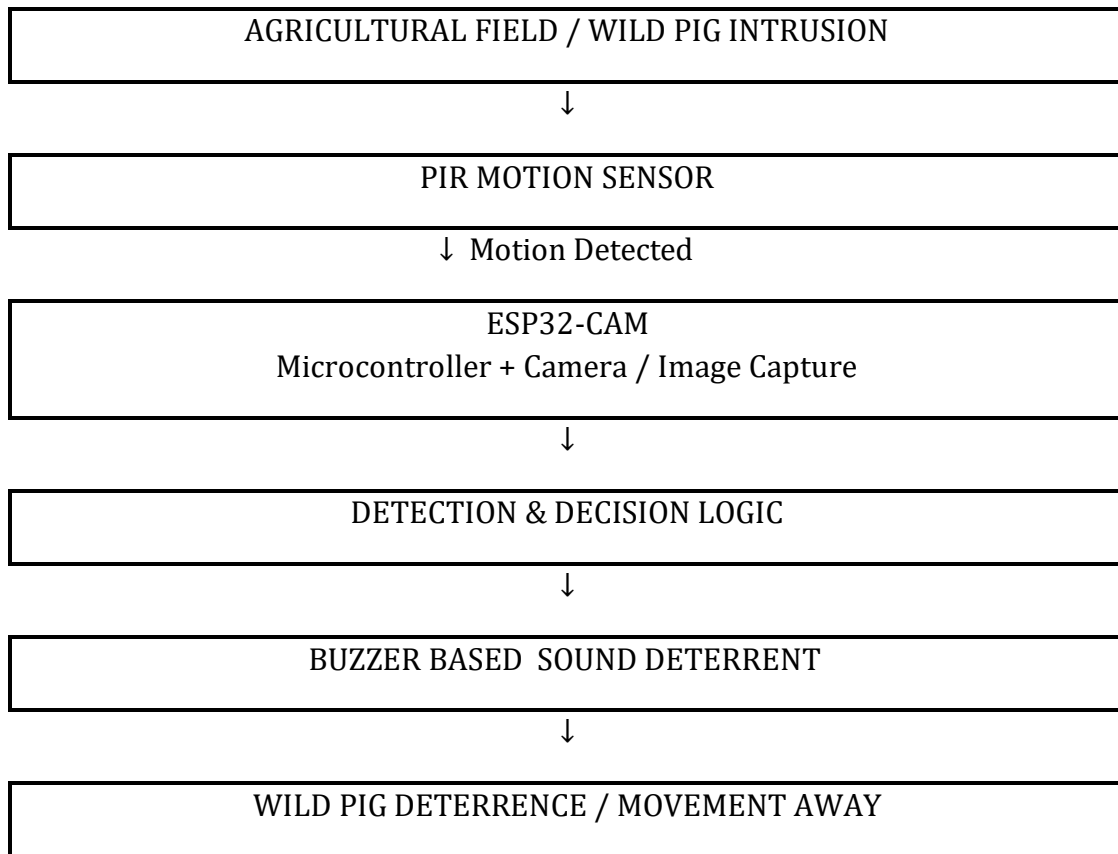
❖ **Software Design:**

- Flowcharts
- Algorithms
- Communication Protocols
- Database Structure (if applicable)

I-SYSTEM ARCHITECTURE

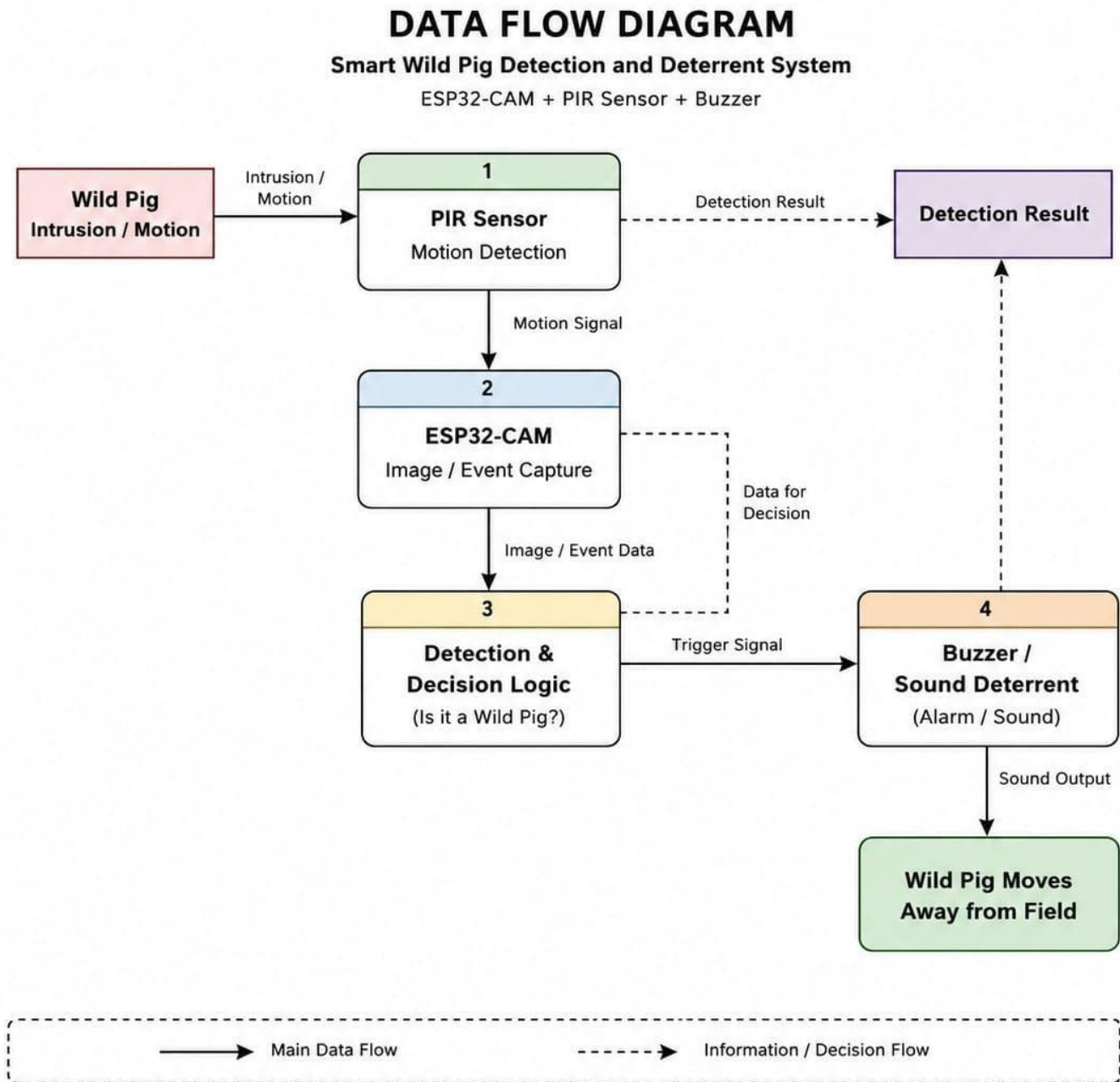
(1).OVERALL SOLUTION ARCHITECTURE:

The proposed system is designed for automatic detection and deterrence of wild pig intrusion in agricultural fields. It combines motion sensing, image-based verification, decision logic, and an ultrasonic deterrent to provide continuous field monitoring.



(2) DATA FLOW DIAGRAM:

Smart Wild Pig Detection and Deterrent System using ESP32-CAM, PIR Sensor and Ultrasonic Sound



- Data Flow: Motion → Sensor Signal → Image/Event Data → Decision → Trigger Signal → Deterrence.
- Wild pig movement is detected by the PIR sensor.
- The PIR sensor sends a motion signal to the ESP32-CAM.
- The ESP32-CAM captures the image/event data.
- The Detection & Decision Logic determines the required action.
- A trigger signal activates the buzzer.
- The buzzer produces a sound to deter the wild pig

(3) HARDWARE – SOFTWARE PARTIONING:

HARDWARE PARTITION:

Hardware Component	Hardware Function
ESP32-CAM	Main controller and camera interface
PIR Sensor	Detects motion/intrusion
Camera	Captures images of detected events
Ultrasonic Sound Module	Produces deterrent sound
Power Supply/Battery	Supplies power to the system
Voltage Regulation	Provides appropriate supply voltage where required
Wi-Fi Hardware in ESP32	Provides wireless connectivity

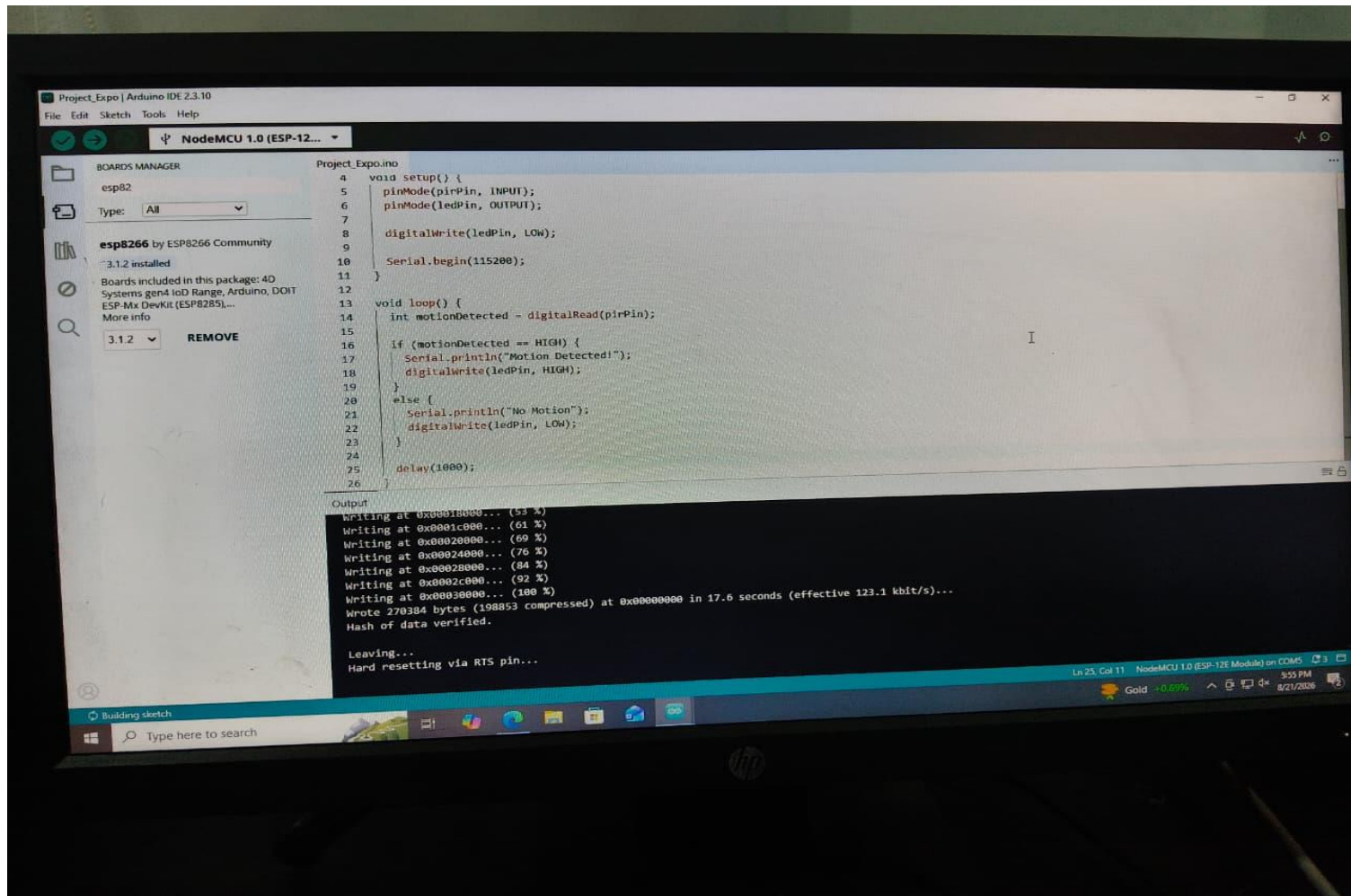
SOFTWARE PARTITION:

Sensor Monitoring	Continuously monitors the PIR sensor
Event Triggering	Activates the camera when motion is detected
Image Capture	Captures the intrusion image/event
Decision Logic	Determines the appropriate system response
Deterrent Control	Controls the ultrasonic module
Wi-Fi Communication	Connects the ESP32-CAM to the network
IoT Alert Generation	Sends intrusion information to the monitoring platform
Notification Handling	Provides an alert to the farmer
Data Logging	Stores event information if implemented

SIMULATION :

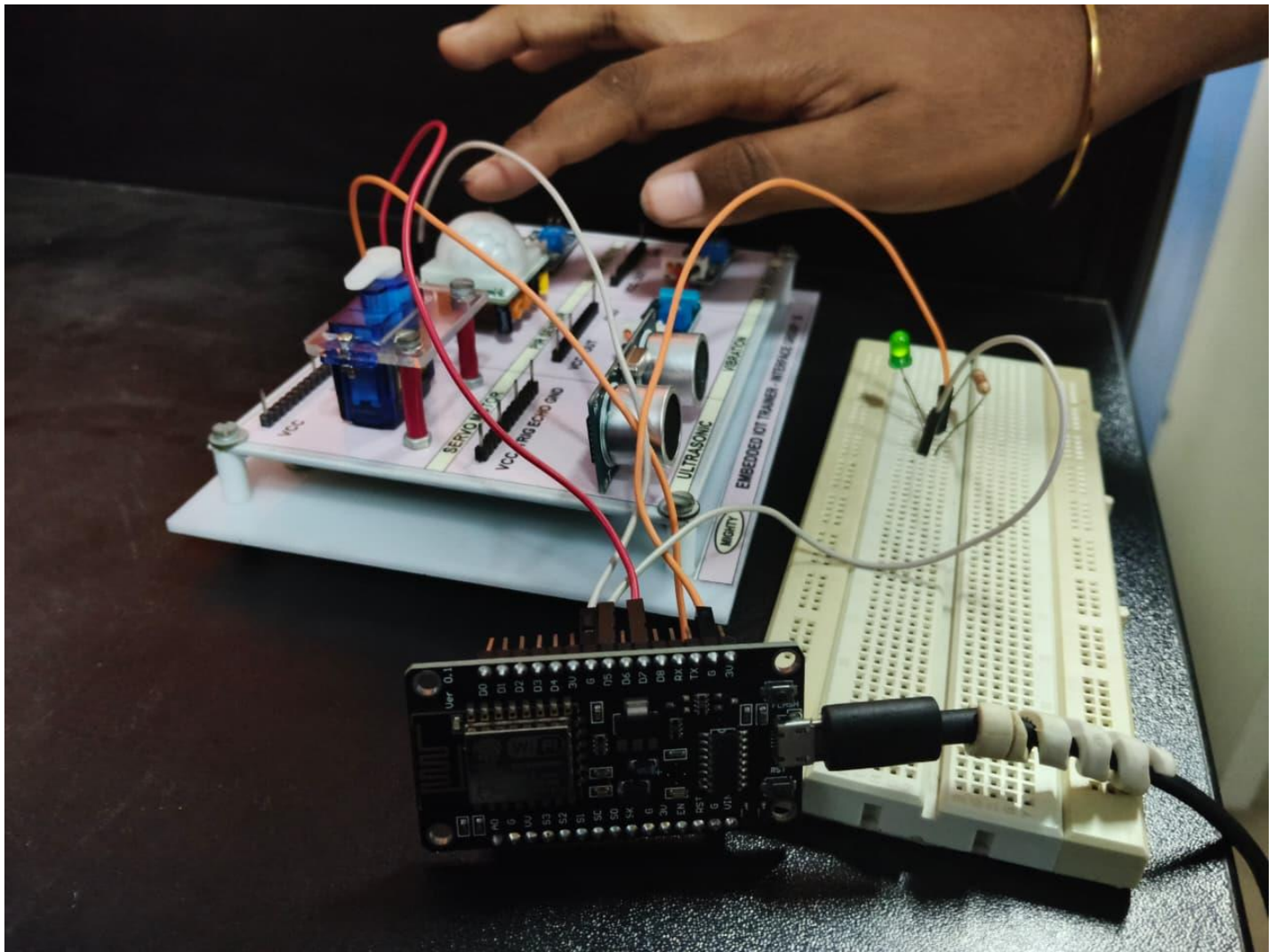
Initial Simulation – Motion Detection

Simulation Completed Using PIR Sensor, LED and ESP8266 NodeMCU 1.0



Arduino IDE Programming & Testing

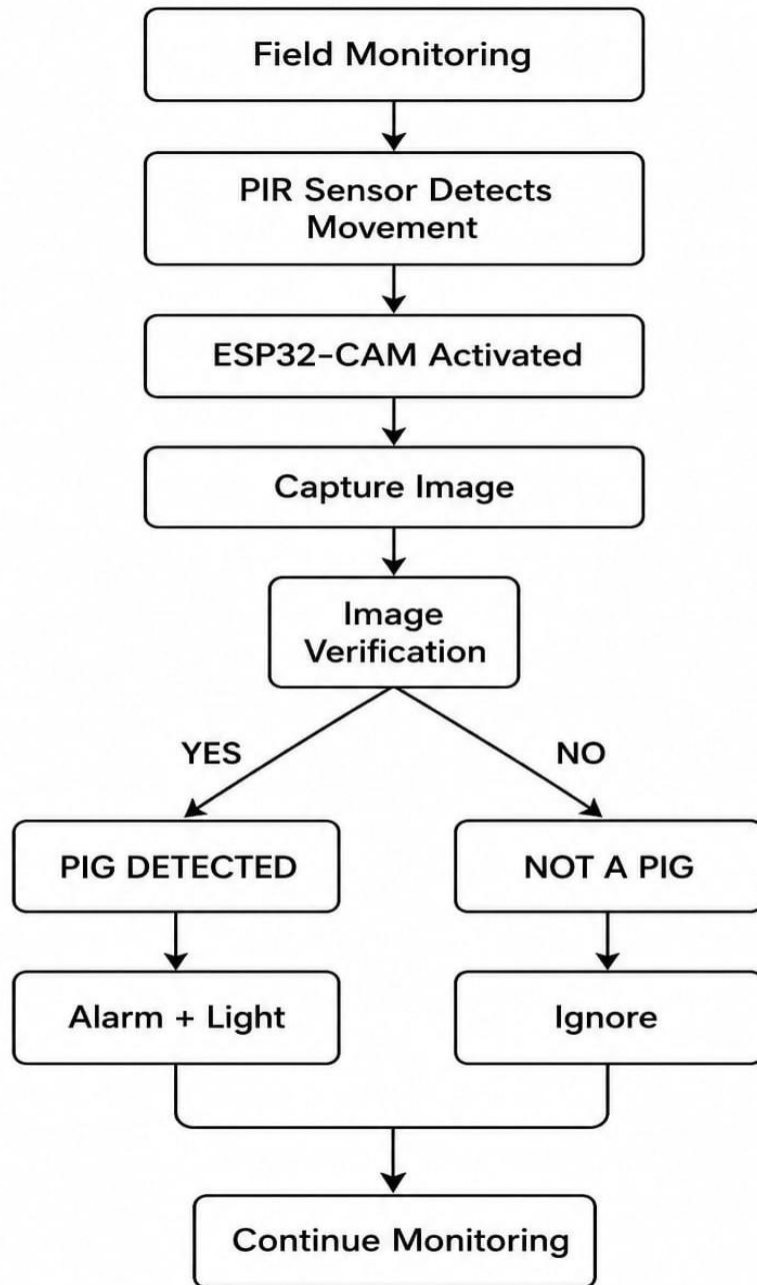
- The **ESP8266 NodeMCU 1.0** was selected and programmed using **Arduino IDE**.
- The program reads the output of the **PIR sensor** continuously.
- When motion is detected, the Serial Monitor displays **“Motion Detected”** and the LED turns **ON**.
- When there is no motion, **“No Motion”** is displayed and the LED remains **OFF**.
- The program was successfully **compiled and uploaded** to the NodeMCU.
- Successful uploading and hardware reset confirmed that the code was executed correctly.



- Successfully developed the **first stage of the wild pig detection system**.
- An **PIR sensor** is used to detect movement in the monitored area.
- The **ESP8266 NodeMCU 1.0** acts as the main controller and processes the PIR sensor output.
- When motion is detected, the NodeMCU activates the **LED as a visual alert**.
- When no motion is detected, the LED remains **OFF** and the system continues monitoring.
- This simulation verifies the basic **motion detection and alert mechanism** of our proposed system.
- This will serve as the **first stage** before integrating the **ESP32-CAM for image-based pig verification**.

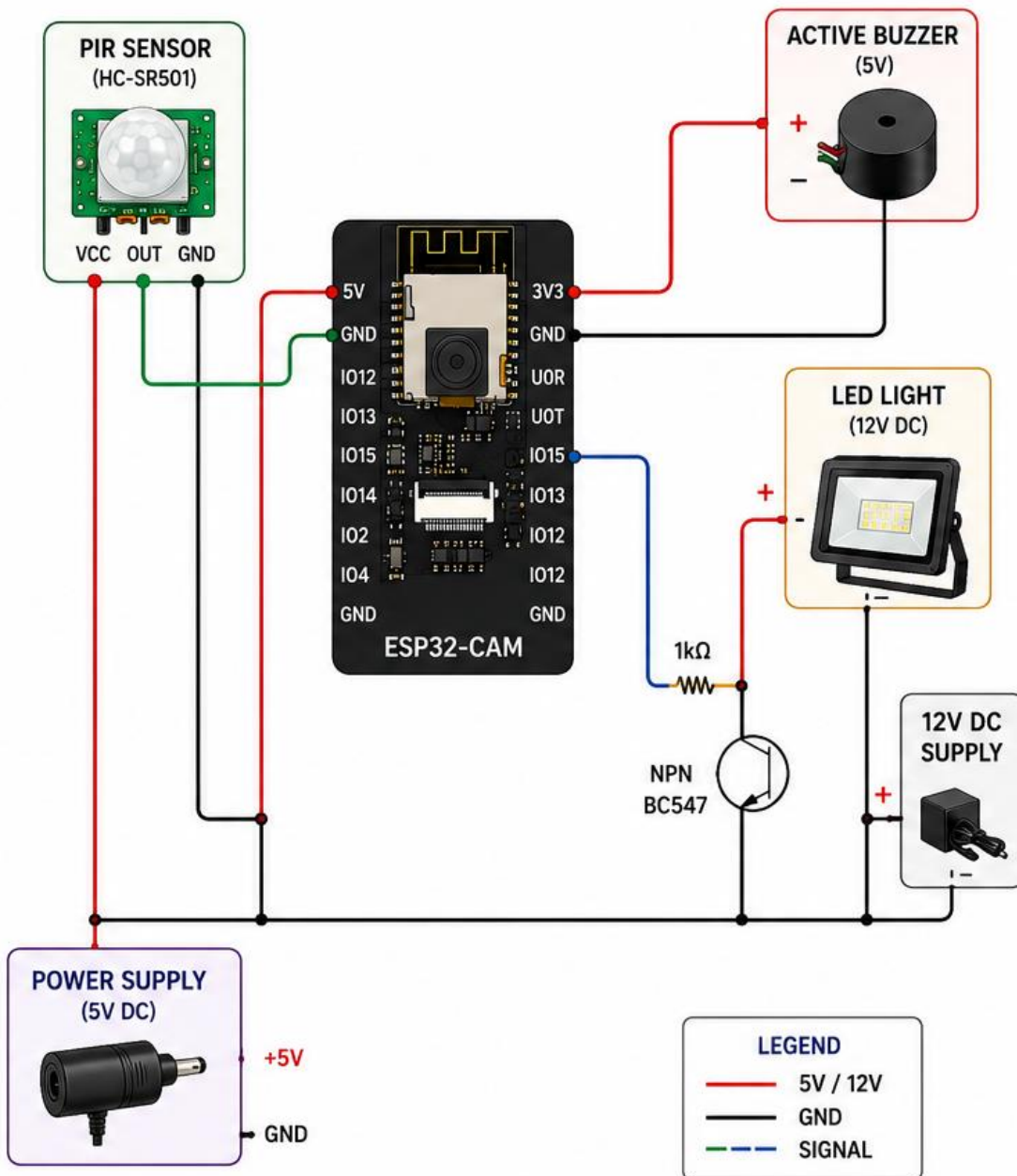
II - HARDWARE DESIGN

(1) FINAL BLOCK DIAGRAM:



(2) CIRCUIT SCHEMATICS:

CIRCUIT SCHEMATICS



Power Supply and Grounding:

The ESP32-CAM requires a **regulated 5V input** to the 5V pin for stable operation, especially when using the camera and Wi-Fi simultaneously. The **3.3V pin** is derived from the onboard regulator and can power small external sensors, but the OV2640 camera must be powered via its **3.3V pin**, never directly from 5V, to prevent damage. All grounds (GND) must be common across the board, camera, and peripherals to avoid signal noise. Adding a **100nF decoupling capacitor** at the camera's power input helps suppress voltage spikes that can cause image artifacts, and a **500mA PTC fuse** is recommended if powering via USB to prevent brownouts during initialization.

Camera Interface

The OV2640 camera connects to the ESP32-CAM through a **24-pin FPC connector**. Key signal connections include:

- **SIOC (I2C clock)** → GPIO 22
- **SIOD (I2C data)** → GPIO 21
- **VSYNC** → GPIO 25
- **HREF** → GPIO 23
- **PCLK** → GPIO 22
- **XCLK** → GPIO 0

These signals handle image synchronization, pixel clock, and camera control. Short wires (under 8 cm) and proper routing prevent interference and data corruption.

Flash LED and Reset

The onboard or external **flash LED** connects to **GPIO 4** with a **220Ω current-limiting resistor** to prevent overheating. The **reset button** is wired to **GPIO 0** with a **10kΩ pull-up resistor**; grounding this pin triggers a restart, while it must remain HIGH during normal operation.

UART and Programming

The ESP32-CAM lacks a built-in programmer, so **UART pins** are used for uploading code:

- **TX** → GPIO 1
- **RX** → GPIO 3

A **3.3V USB-serial adapter** is required; using a 5V adapter without level shifting can damage the board. The FTDI programmer is commonly used for this purpose.

microSD Card Interface

The microSD card slot allows image and video storage. Pins used include **SD_DATA0–3** and **SD_CLK**. Cards should be formatted as **FAT32**, and using larger clusters (32KB+) improves sequential write performance. Avoid using SD pins for other peripherals simultaneously to prevent conflicts.

Antenna and Wi-Fi :

The ESP32-CAM has a **built-in PCB trace antenna**. For better Wi-Fi performance, a **U.FL connector** can be used to attach a 2.4GHz dipole antenna. Enclosures, especially metal, can reduce signal strength or create dead zones, so testing without enclosures is recommended .

GPIO Overview :

The board has limited GPIOs, many of which are **dual-purpose**:

- Some pins are **boot-sensitive** (e.g., GPIO 0, 2, 15) and affect flash mode.
- Others serve as **camera signals, SD card lines, or general I/O**.
- Careful attention to pin functions avoids conflicts during programming and operation.

(3) COMPONENT SELECTION:

- ❖ The components were selected based on their cost, power consumption, ease of interfacing, and suitability for agricultural field conditions.
- ❖ The PIR sensor (HC-SR501) was selected because it is low-cost, consumes less power, and can easily detect movement.
- ❖ The ESP32-CAM was selected because it is compact, low-cost, and has a built-in camera for capturing images after motion is detected.
- ❖ An active buzzer was chosen because it is inexpensive and easy to interface, providing an audible alarm when a pig is detected.
- ❖ An LED light or floodlight was selected to provide visual deterrence, especially during nighttime.
- ❖ A 5 V DC power supply was selected because it is compatible with the ESP32-CAM and PIR sensor.

Component	Selected Component	Purpose	Reason for Selection

Motion Sensor	PIR Sensor (HC-SR501)	Detects movement in the field	Low cost, low power consumption and easy to use
Camera & Controller	ESP32-CAM	Captures images after motion detection	Compact, low-cost with built-in camera
Alarm	Active Buzzer	Produces a sound when a pig is detected	Low cost and easy to interface
Light	LED Light / Floodlight	Activates to deter wild pigs, especially at night	Suitable for visual deterrence
Power Supply	5 V DC Supply	Powers ESP32-CAM and PIR sensor	Compatible with the system

(4) BILL OF MATERIALS(BOM) :

MAJESTRONICZ

10, Nachiappa 2nd St,
Nachiyappa Colony, Kottai,
Erode, Tamil Nadu 638001

☎ 6379560289

Company GST :
33ABZFM5739L1ZD
Tamil Nadu



Bill To:
LAVANYA V
BODINAYAKANUR
☎ 6383806229

Ref No:
MZERD26-27/5963
Date of Issue
17-08-2026

SR	Name	Qty	Price	Amount
1	5V -1A ADAPTER HSN/SAC :8504	1Pcs	109.00	109.00
2	ESP32 CAMERA MODULE	1Pcs	699.00	699.00
Total		2		808.00
			Shipping	75.00
			Total	883.00
			Received	883.00

NO WARRANTY **NO EXCHANGE** **NO
RETURN**

The major components required for the prototype were procured from Majestronicz. The ESP32-CAM module and power adapter were purchased for the project. Other supporting components, including the PIR sensor, LED, buzzer, breadboard, and jumper wires, were available in the college laboratory and were used for prototype development

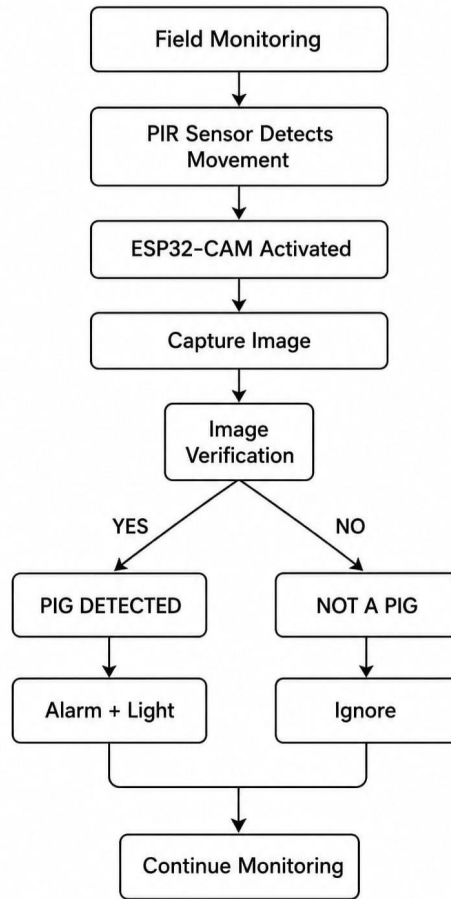
III - Software design

The software is designed to control the ESP32-based monitoring system, process the PIR sensor signal, capture images using the ESP32-CAM, verify the detected animal, and activate an alert when a wild pig is identified.

(1)-Software Components :

Component ID	Software Component	Function
SC-01	Sensor Monitoring Module	Continuously monitors the PIR sensor and detects movement in the field.
SC-02	ESP32 Controller Module	Receives the PIR signal and controls the overall system operation.
SC-03	Camera Control Module	Controls the ESP32-CAM and captures an image when motion is detected.
SC-04	Image Verification Module	Analyzes the captured image to determine whether the detected animal is a wild pig.
SC-05	Decision Module	Makes the decision to activate the alert when a wild pig is confirmed.
SC-06	Alert Control Module	Activates the alarm and ultrasonic deterrent when a wild pig is detected.
SC-07	Monitoring Module	Returns the system to normal monitoring after the alert operation.
SC-08	Communication Module	Provides Wi-Fi/IoT communication for remote monitoring, if implemented.
SC-09	Data Storage Module	Optionally stores detection records such as date, time, result, and image reference.

FLOWCHART:



(2)-ALGORITHM:

- Step 1: Start the system.
- Step 2: Initialize ESP32, PIR sensor and ESP32-CAM.
- Step 3: Continuously monitor the PIR sensor.
- Step 4: If no movement is detected, continue monitoring.
- Step 5: If movement is detected, trigger the ESP32-CAM.
- Step 6: Capture an image of the detected animal.
- Step 7: Perform image verification.
- Step 8: If the animal is not a wild pig, ignore the detection and return to monitoring.
- Step 9: If the animal is a wild pig, activate the alarm and ultrasonic deterrent.
- Step 10: Return to monitoring mode.
- Step 11: Repeat continuously.

(3)-COMMUNICATION PROTOCOLS:

Protocol ID	Communication Protocol	Used For	Function
CP-01	GPIO	PIR Sensor ↔ ESP32	Transfers the motion detection signal from the PIR sensor to the ESP32.
CP-02	GPIO	ESP32 ↔ Alarm/Ultrasonic Device	Sends control signals to activate the alarm and ultrasonic deterrent.
CP-03	Wi-Fi	ESP32 ↔ Network	Provides wireless connectivity for IoT-based monitoring.
CP-04	HTTP	ESP32 ↔ Server/Application	Can be used to send detection information or receive commands through a network.
CP-05	UART/Serial	ESP32 ↔ Computer	Used for programming, debugging, and monitoring system messages during development.

(4)-DATABASE STRUCTURE:

Field	Description
Detection ID	Unique detection number
Date	Date of detection
Time	Time of detection
Animal	Detected animal
Confidence	Detection confidence
Image	Captured image reference
Alert_Status	Alert activated or not